RouteRec: Behavior-Guided Expert Routing for Sequential Recommendation

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Abstract

Sequential recommendation is commonly modeled with a largely shared computation pipeline, even though user behavior can vary substantially across sessions and examples. Some interactions reflect stable preference, while others are shaped more by repetition, popularity sensitivity, or short-term exploration. This heterogeneity suggests that different cases may benefit from different computation paths, but existing expert-based approaches often rely on hidden-state-driven routing, making specialization harder to interpret and control.

We propose RouteRec, a feature-guided mixture-of-experts framework for sequential recommendation. RouteRec infers compact behavioral cues from simple and broadly available sequential signals and uses them exclusively for routing, while expert networks transform sequence hidden states. By separating why an example is routed from how its representation is transformed, RouteRec makes expert specialization more explicitly grounded in user and session characteristics.

Experiments show that RouteRec consistently improves next-item recommendation over strong baselines. Overall, the results suggest that lightweight cues from standard interaction logs can provide effective and interpretable routing signals for adaptive computation in sequential recommendation.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

sequential recommendation, mixture of experts, conditional computation, routing, behavioral features

1 Introduction

Sequential recommendation is commonly formulated as next-item prediction from an ordered interaction history. While the prediction objective is simple, the behavioral structure behind it is often not. A user may exhibit stable long-term preference in one context, repeated consumption in another, and short-term exploratory behavior near the current prediction point. Even within the same user, the dominant pattern can shift across sessions and over time. This heterogeneity suggests that different user–session contexts may not always benefit from the same computation path, which is also aligned with prior survey views on sequential and session-based recommendation [26, 27].

Despite this, most sequential recommenders still rely on largely shared computation. Modern sequence encoders have become increasingly strong at modeling transitions and long-range dependencies, but they typically apply the same transformation pipeline to behaviorally different cases [5, 8, 12, 15, 18, 20]. This shared-computation design is effective and efficient, yet it can also blur distinctions among examples that would ideally activate different forms of processing. As illustrated in Figure 1, users and sessions may differ not only in broad preference, but also in repetition tendency, popularity sensitivity, and local browsing dynamics. A single shared path can still produce good average predictions, but it does not explicitly explain why different behavioral regimes should trigger different computations.

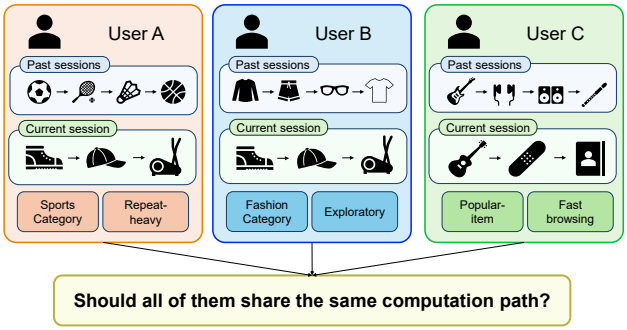

Figure 1: Motivating example of behavior heterogeneity in sequential recommendation. Different users and sessions can exhibit different behavioral characteristics, suggesting that they need not share exactly the same computation path. RouteRec addresses this by extracting compact behavioral cues from simple sequential signals and using them as explicit routing signals.

Mixture-of-experts (MoE) offers a natural mechanism for adaptive computation in this setting. By allocating different inputs to different experts, MoE can in principle specialize model capacity according to behavioral variation [4, 6, 7, 19]. However, simply inserting MoE into sequential recommendation does not by itself resolve the core issue. In many expert-based architectures, routing is driven primarily by hidden states. This makes the basis of specialization largely implicit: the same latent representation that summarizes behavior is also expected to decide how computation should be allocated. As a result, expert assignment may reflect entangled hidden-state geometry rather than clearly identifiable behavioral factors. Such routing can be effective, but it is harder to interpret, harder to diagnose, and harder to control when one wants specialization to correspond to meaningful user or session characteristics.

Our starting point is that many behavior regimes in sequential recommendation are not purely latent. They can

often be described, at least coarsely, by compact cues derived from ordinary interaction logs: whether recent behavior is repeat-heavy, whether the current session is locally exploratory, whether interactions lean toward popular items, or whether the broader history indicates a persistent preference pattern. These cues are lightweight, broadly available, and semantically easier to inspect than hidden-state geometry alone. This motivates a different design principle: instead of asking hidden states to both represent content and determine routing, use explicit behavioral cues to control routing and reserve hidden states for expert transformation.

Based on this idea, we propose **RouteRec**, a behavior-guided mixture-of-experts framework for sequential recommendation. RouteRec derives compact behavioral cues from simple sequential signals, including interaction order, timestamps, and optional coarse item categories, and uses them exclusively for routing. Expert networks, in turn, operate on sequence hidden states. This role separation makes the routing decision more explicitly grounded in user and session characteristics: behavioral cues explain *why* an example is routed to certain experts, while hidden representations determine *how* those experts transform the sequence state. In this sense, RouteRec is not merely a MoE layer added to a sequential backbone, but a routing design that aims to make specialization behavior-aware by construction.

A further challenge is that behavioral variation does not appear at a single temporal scale. Some signals reflect broad cross-session tendency, some describe evolving within-session intent, and others arise only from the most recent transitions near the prediction target. RouteRec therefore organizes routing across multiple temporal scopes, allowing expert selection to respond to both persistent and short-lived behavioral evidence while keeping the main representation pipeline simple.

The main contributions of this paper are as follows. First, we introduce RouteRec, a behavior-guided MoE framework that addresses computation-path heterogeneity in sequential recommendation. Second, we construct a compact cue bank from simple sequential signals and use it exclusively for routing rather than direct representation enrichment. Third, we show empirically that explicit behavior-guided routing improves recommendation performance over strong baselines and yields more interpretable specialization behavior.

2 Related Work

2.1 Sequential Recommendation with Shared Computation

Sequential recommendation has evolved through a range of backbone architectures, including recurrent, convolutional, graph-based, and Transformer-based models [5, 8, 21, 23, 28]. Early methods focused on modeling short-term transitions and session intent more effectively [11, 14], while later approaches established self-attention as a dominant backbone for capturing both local and long-range dependencies [8, 21]. More recent work has further improved sequence modeling through stronger training objectives, contrastive regularization, and structural inductive bias [3, 12, 15, 18, 20, 24, 29]. Although these models differ substantially in architecture and optimization, they mostly retain a common design principle: behaviorally different examples are processed through largely shared transformations. RouteRec is related to this line in that it builds on a strong sequential backbone, but it departs from it by explicitly treating computation-path heterogeneity as a modeling target.

2.2 Modeling Heterogeneous User and Session States

Another line of research recognizes that user behavior is not governed by a single stationary interest. Session-based and sequential recommendation studies have explored short-term intent, long- versus short-term preference, repeated consumption, and multi-interest structure [2, 10, 11, 14, 16, 31]. These methods show that user behavior can vary across sessions and time scales, and that capturing such variation is important for prediction quality. However, most of them address heterogeneity at the representation level, for example by designing better encoders, attention mechanisms, or interest disentanglement modules. In contrast, RouteRec focuses on heterogeneity from the perspective of conditional computation: rather than only refining the representation itself, it asks whether different behavioral regimes should activate different expert pathways.

2.3 Feature-Aware and Time-Aware Sequential Recommendation

Feature-aware sequential models incorporate auxiliary signals such as timestamps, intervals, side information, or feature channels to improve sequence modeling [3, 12, 20, 30, 33]. This line of work has shown that even lightweight contextual signals can complement item transitions and strengthen recommendation accuracy. In most cases, however, these signals are used to enrich input representations, bias attention, or support feature fusion within a shared computation pipeline. RouteRec is motivated by a different use of such signals. Instead of treating behavioral features primarily as representation enrichers, it uses compact cues derived from simple sequential logs as explicit routing signals for expert selection.

2.4 Mixture-of-Experts and Conditional Computation

Mixture-of-experts and conditional-computation models provide the broader perspective behind RouteRec. Classical expert mixtures introduced the idea of allocating different inputs to specialized subnetworks [6, 7], and modern sparse MoE models demonstrated that conditional capacity can scale effectively when routing is informative and numerically stable [4, 9, 19, 25, 35]. Related ideas have also appeared in recommendation, including multi-expert decomposition and expert-based personalization [1, 13, 17, 22, 32]. These studies establish the usefulness of expert specialization, but in many settings routing remains primarily a function of hidden states or task-specific latent representations. RouteRec

differs in routing semantics: it uses compact behavioral cues from ordinary interaction logs as the primary control signal for stage-wise expert selection, with the goal of making specialization more explicit, inspectable, and behavior-aware.

3 Methodology

3.1 Problem Setting and RouteRec Overview

Sequential recommendation predicts the next item from an ordered interaction prefix. We organize each user history into sessions and omit the user index for readability. Let $\mathcal{S} = [s_1, \ldots, s_M]$ denote the session sequence, and let $s_m = [x_1, \ldots, x_{|s_m|}]$ denote the current session. Each interaction x_t contains at least item identity and timestamp, and may include a coarse category label. Given observations up to position t, the model estimates the next-item distribution.

RouteRec keeps the SASRec backbone unchanged at the representation level [8], but replaces shared feed-forward transformation with stage-wise MoE routing [4, 7, 19]. The architectural principle is explicit role separation: stage features determine routing preference, while experts transform hidden representations. This separation allows the model to keep a strong common encoder while still allocating computational capacity according to behavioral context.

Let $\mathcal{T} = \{\text{macro}, \text{mid}, \text{micro}\}$ be the stage set, $\mathbf{h}_t$ be the hidden state at position t, and $\mathbf{f}_t^{(s)}$ be the stage feature used by router s. Route selection and expert aggregation are written as

$$\boldsymbol{\pi}_t^{(s)} = \mathrm{softmax}\Big(\mathrm{Router}_s(\mathbf{f}_t^{(s)})/\tau_s\Big), \tag{1}$$

$$\tilde{\mathbf{h}}_t^{(s)} = \sum_{e=1}^{E} \pi_{t,e}^{(s)} \, \mathrm{Expert}_e^{(s)}(\mathbf{h}_t). \tag{2}$$

Equation (2) is shared across stages, but temporal meaning and routing structure differ. Macro routing captures broad cross-session tendency, mid routing adapts to evolving within-session context, and micro routing reacts to immediate transition signals. In this way, RouteRec aligns routing granularity with temporal scope instead of relying on a single fixed specialization rule.

3.2 Behavioral Features from Raw Signals

RouteRec is designed for realistic data availability. It uses signals that are commonly present in interaction logs: item identity, timestamp, optional category, and optional corpus-level popularity. We intentionally avoid heavy side-information encoders in the routing branch. Instead, raw signals are converted into compact scalar summaries that are easy to compute, stable across datasets, and interpretable at analysis time.

For session s_m and target position t, the stage windows are

$$\mathcal{R}_t^{\text{macro}} = [s_{\max(1,m-H)}, \ldots, s_{m-1}], \tag{3}$$

$$\mathcal{R}_t^{\text{mid}} = [x_1, \ldots, x_{t-1}], \tag{4}$$

$$\mathcal{R}_t^{\text{micro}} = [x_{\max(1,t-w)}, \ldots, x_{t-1}], \tag{5}$$

where $H = 5$ and $w = 5$ in our main setting. A shared summary operator $\mathcal{A}(\cdot)$ maps each window to a low-dimensional feature vector:

$$\mathbf{f}_t^{(s)} = \mathcal{A}(\mathcal{R}_t^{(s)}), \qquad s \in \mathcal{T}.$$

The summary dimensions are grouped into four semantic families: *Tempo*, *Focus*, *Memory*, and *Exposure*. Tempo captures pace and interval statistics. Focus captures concentration and switching behavior over coarse categories or item groups. Memory captures recurrence, repetition, and overlap with recent context. Exposure captures interactions with popularity strata and their local drift. This decomposition is semantic rather than architectural: it gives routing modules an interpretable behavioral basis.

Importantly, the feature template is shared across stages, but each stage reads it through a different temporal lens. Macro features reflect persistent tendency and broad preference shape. Mid features track evolving intent inside the current session. Micro features emphasize immediate cues around the prediction target. This shared-template, stage-specific-view design keeps the feature branch compact while preserving temporal specificity.

3.3 Feature-Guided Multi-Stage Routing

The same routing interface is used at all stages, but each stage adopts a different routing bias to match its temporal evidence quality.

Macro stage (group-first). Macro features are comparatively stable within one session and are better suited for coarse specialization than fine-grained instant decisions. We therefore use hierarchical group-first routing. Let $g(e)$ be the group index of expert e. Group logits and within-group logits are added in logit space:

$$q_{t,e}^{\text{macro}} = q_{t,g(e)}^{\text{grp}} + q_{t,e|g(e)}^{\text{intra}}.$$

This first selects a broad specialization regime and then refines expert preference inside that regime, which is a natural match for long-horizon behavioral evidence.

Mid stage (hybrid). Mid features evolve at every position and can deviate from long-range tendency while still preserving meaningful structure. To reflect this, the mid router combines structured and direct branches:

$$\mathbf{q}_t^{\text{mid}} = \alpha \, \mathbf{q}_t^{\text{struct}} + (1-\alpha) \, \mathbf{q}_t^{\text{direct}},$$

where $\alpha \in [0, 1]$ balances regularity and flexibility. The structured branch retains group-aware prior, while the direct branch allows fast correction when current session dynamics diverge from that prior.

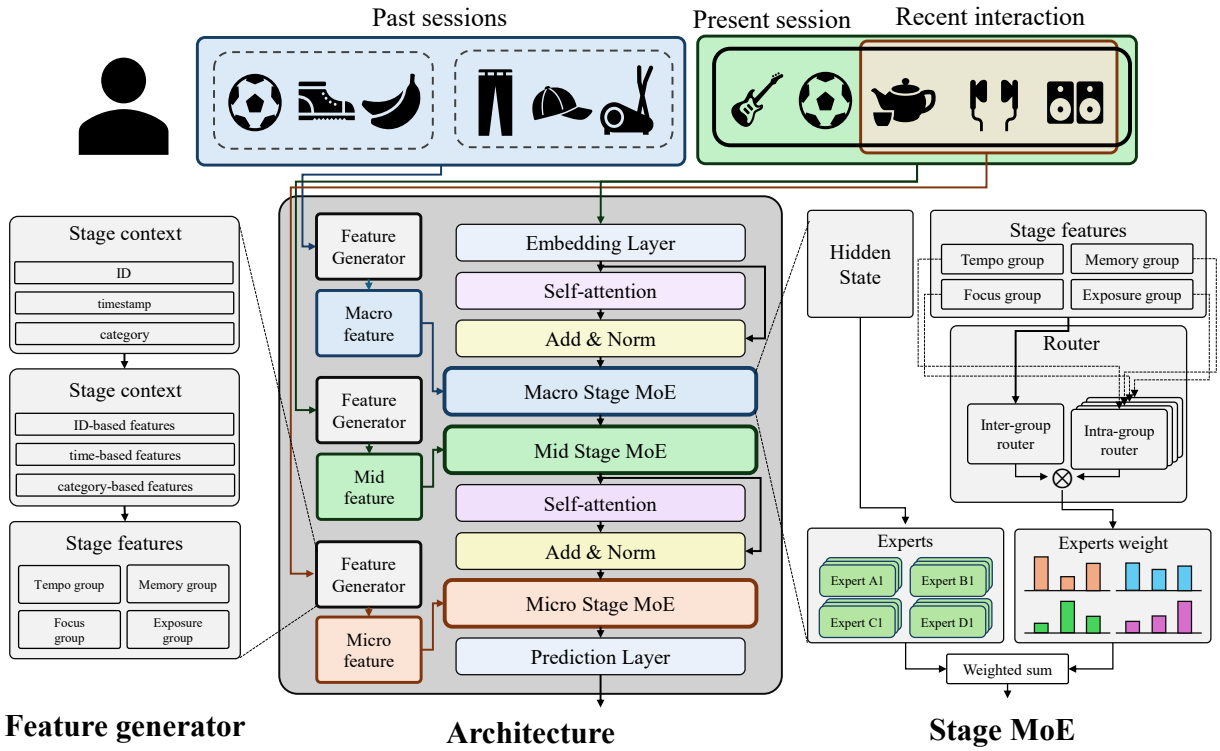

Figure 2: Overall RouteRec framework. (a) RouteRec keeps the SASRec backbone and replaces shared feed-forward computation with stage-wise mixture-of-experts routing; stage features determine routing weights, while expert FFNs transform hidden states. (b) The routing computation is built from three reusable components: group-level routing, conditional within-group expert routing, and direct expert routing. (c) The three stages combine these components differently: the macro stage uses hierarchical routing, the mid stage blends hierarchical and direct routing, and the micro stage uses direct expert routing.

Micro stage (direct). Micro features are highly local and closest to the prediction target. At this scope, forcing an additional group bottleneck is often unnecessary, so direct routing is used:

$$\mathbf{q}_t^{\text{micro}} = \mathbf{q}_t^{\text{direct}}.$$

Routing weights follow the softmax rule in Eq. (2). Overall, RouteRec forms a coarse-to-fine routing pipeline in which routing structure is explicitly synchronized with temporal scope.

3.4 Training Objective

RouteRec is trained with a standard next-item objective and two router-side regularizers: route consistency and z-loss. The first regularizer shapes routing semantics, and the second stabilizes routing numerics.

Let $\mathbf{z}$ be final logits and y be the target item. The main loss is

$$\mathcal{L}_{\text{CE}} = -\log \frac{\exp z_y}{\sum_j \exp z_j}.$$

For route consistency, we first compute a sample-level routing distribution at each stage by averaging token-level routing weights:

$$\bar{\boldsymbol{\pi}}_i^{\,s} = \frac{1}{L_i} \sum_{t=1}^{L_i} \boldsymbol{\pi}_{i,t}^{\,s}$$

where L_i is the valid sequence length. For each stage s, we construct a pair set $\mathcal{P}_s$ from feature-similar samples in the mini-batch and minimize

$$\mathcal{L}_{\text{cons}} = \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} \frac{1}{|\mathcal{P}_s|} \sum_{(i,j) \in \mathcal{P}_s} \text{JSD}\big(\bar{\boldsymbol{\pi}}_i^{\,s}, \bar{\boldsymbol{\pi}}_j^{\,s}\big),$$

which encourages neighboring behavioral states to produce compatible routing distributions. In other words, semantically similar inputs should not induce arbitrarily different routing behavior.

To stabilize gating numerics, we adopt z-loss on router logits [36]:

$$\mathcal{L}_z = \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} \frac{1}{N_s} \sum_{(i,t) \in \mathcal{V}_s} \left(\log \sum_{e=1}^{E} \exp q_{i,t,e}^{\,s} \right)^2,$$

where $\mathcal{V}_s$ is the set of valid positions and $N_s = |\mathcal{V}_s|$. This term discourages excessive logit scale and helps prevent unstable,

overly sharp routing. The final objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{cons}} \mathcal{L}_{\mathrm{cons}} + \lambda_z \mathcal{L}_z,$$

where λ_{cons} and λ_z are trade-off coefficients. This objective jointly optimizes prediction quality, semantic consistency of routing, and numerical stability of expert selection.

4 Experiments

4.1 Experimental Setup

Our main model follows Section 3: macro–mid–micro stage composition, hierarchical macro routing, hybrid mid routing, direct micro routing, feature-guided learned routers, and no bias-specific routing terms. Training uses next-item cross-entropy with route consistency and z-loss, as defined in Section 3.4. Unless otherwise stated, all reported RouteRec results use this unified configuration.

We compare RouteRec against representative recurrent, graph-based, Transformer-based, frequency-aware, Mamba-based, and MoE-style sequential recommenders. The primary baseline set includes GRU4Rec, SASRec, TiSASRec, and DuoRec [5, 8, 12, 18], plus BSARec, SIGMA, DIF-SR, and FAME [13, 15, 20, 30]. When budget and space allow, we additionally include Caser, SR-GNN, BERT4Rec, FMLP-Rec, and FEARec [3, 21, 23, 28, 34] to broaden architectural coverage.

For fairness, we prioritize a unified training framework: matched embedding dimension, hidden dimension, optimizer family, batch regime, and early-stopping protocol wherever implementation permits. Data preprocessing is documented with sessionization policy, filtering thresholds, sequence truncation rules, and feature availability (timestamp/category/popularity). Evaluation follows standard top-K ranking metrics, including HR@K, NDCG@K, and MRR@K. We report MRR@20 as the primary summary metric and present other metrics for completeness.

4.2 Main Results

This subsection provides the overall performance comparison between RouteRec and all baselines across target datasets. The main table reports ranking metrics at common cutoffs and highlights RouteRec gains over strong Transformer and MoE references.

We focus analysis on three questions. First, does feature-guided routing improve over a strong shared-backbone baseline such as SASRec under matched training conditions? Second, are gains consistent across datasets with different temporal density and behavioral regularity? Third, does RouteRec remain competitive when compared with recent feature-aware or MoE-style alternatives rather than only classical baselines?

Beyond absolute ranking numbers, we summarize relative improvements against SASRec and the strongest non-RouteRec competitor on each dataset. This framing clarifies whether improvements are robust or concentrated in a narrow subset of settings.

4.3 Ablation and Diagnostic Analysis

This subsection analyzes why RouteRec works through controlled ablations and routing diagnostics. We first evaluate router-design variants (hierarchical, hybrid, direct combinations) to test whether stage-specific routing bias is necessary. We then evaluate feature-family ablations to measure the contribution of Tempo, Focus, Memory, and Exposure cues.

Next, we examine stage composition by removing or altering macro, mid, and micro branches, verifying whether coarse-to-fine routing is materially better than a single-stage or uniformly structured alternative. We also include corrupted-feature sanity checks to confirm that gains are tied to meaningful behavioral cues rather than incidental regularization effects.

Finally, we present routing diagnostics, including expert usage distribution, stage-wise route entropy, and consistency behavior under similar feature states. These analyses connect quantitative improvements to the intended mechanism: behavior-aware specialization rather than opaque capacity inflation.

5 Conclusion

The conclusion will restate the central claim of the paper: compact behavioral cues can serve as explicit routing guidance for mixture-of-experts sequential recommendation. It will summarize the empirical findings and discuss portability, interpretability, and future extensions of the framework.

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